NANYANG TECHNOLOGICAL UNIVERSITY

SEMESTER 2 EXAMINATION 2020-2021

CE2005/CZ2005 – OPERATING SYSTEMS

NOTE: This is an online exam conducted via NTULearn

Total Marks: 30

Apr/May 2021 Time Allowed: 1 hour

INSTRUCTIONS

1. This paper contains SIX questions.
2. Answer **ALL** the questions.
3. This is a closed-book examination.
4. Questions carry equal marks.

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1. Consider a set of processes P0, P1, P2 and P3 with arrival times 0, 2, 4 and 8, respectively. A schedule of these processes is shown in Figure Q1 below, including both CPU as well as I/O bursts.

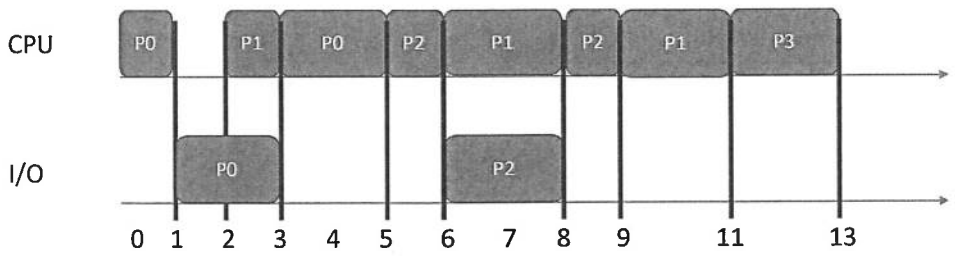


Figure Q1

Identify the CPU and I/O burst lengths for all the processes (if there are multiple bursts, identify all of them). Identify the scheduling algorithm and compute the resulting average waiting time.

(5 marks)

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2. Consider the following implementations of the semaphore system calls **Wait()** and **Signal()**.

```
Wait (Semaphore S) {
  S.value--;
  If (S.value < -1) {
    block();
  }
}
```

```
Signal (Semaphore S) {
  S.value++;
  If (S.value <= 0) {
    wakeup();
  }
}
```

Suppose these system calls are guaranteed to be atomic and a Semaphore S in this system is used to protect critical sections as shown below. Assume S is initialized to 0. Explain if this solution ensures mutual exclusion and progress with justifications.

```
Process Pi:
  While(1){
    Wait(S);
    Critical-section for Pi
    Signal(S);
    Remainder-section for Pi
  }
```

(5 marks)

3. Consider the following snapshot of a system's state, with four processes (P0, P1, P2 and P3) and three resource types (A, B and C). The current Allocation, Need and Available matrices are as shown in Table Q2.

Table Q2

		Available A B C 0 x y
Process	Allocation A B C	Need A B C
P0	2 2 1	0 2 0
P1	1 1 0	2 1 2
P2	1 1 1	2 0 1
P3	1 1 2	4 1 1

Suppose process P3 generates the resource request (A=0 B=1 C=1) next. Compute the minimum values for x and y so that this request can be granted and the resulting system state is safe. Justify your answer and clearly show the algorithm steps.

(5 marks)

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4. Assume an operating system uses paging for memory allocation. The page size is 4K bytes. There are 2^{20} entries in a page table and each entry occupies four bytes.

   (a) What is the logical address format?

   (1.5 marks)

   (b) What is the size of the logical address space?

   (1.5 marks)

   (c) If two-level paging scheme is used, what is the logical address format?

   (2 marks)

5. Consider a system that uses demand paging and the following page reference string: 0, 1, 2, 1, 0, 2, 3, 0. Assume that initially all frames are empty.

   (a) Assuming there are three frames available, how many page faults will LRU page replacement algorithm generate?

   (2.5 marks)

   (b) Assuming the number of frames available is larger than one, what is the minimum number of frames required so that FIFO and LRU page replacement algorithms will generate the same number of page faults for the given page reference string? Justify your answer.

   (2.5 marks)

6. Consider a file system on a disk that has both logical and physical block sizes of 512 bytes. Assume that the current block of a file being accessed is block 5 and the directory entry of the file is already in the memory. For each of the three file allocation methods (i.e., contiguous, linked, and indexed), how many physical blocks must be read from the disk if the next block to be accessed is block 10?

   (5 marks)

END OF PAPER

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